IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 26 July 2025, accepted 10 August 2025, date of publication 12 August 2025, date of current version 20 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3598130

RESEARCH ARTICLE

Patient-Specific Digital Twins for Personalized Healthcare: A Hybrid AI and Simulation-Based Framework

HARSHIT SHARMA[1] **AND SIMRAN KAUR**[2]
[1]Department of CSE, MNIT Jaipur, Jaipur, Rajasthan 302017, India
[2]Department of AIML, Manipal University Jaipur, Jaipur, Rajasthan 303007, India

Corresponding author: Simran Kaur (simran.kaur@jaipur.manipal.edu)

This work was supported by the Manipal University Jaipur.

ABSTRACT Digital twins (DTs) represent a transformative paradigm in personalized medicine, enabling real-time, patient-specific simulations that support precision diagnosis, continuous monitoring, and adaptive treatment planning. This paper presents a novel hybrid framework that integrates mechanistic physiological modeling with deep learning to construct patient-specific digital twins. The proposed architecture couples a dynamic simulation engine—capable of modeling organ-system interactions—with neural encoder-decoder networks that extract latent representations from heterogeneous clinical data sources, including biosignals (e.g., ECG), medical imaging, and wearable sensor streams. A probabilistic learning module further enables predictive adaptation under uncertainty, leveraging Bayesian inference and reinforcement learning. The framework supports continuous synchronization between a patient's real-world physiological state and its digital counterpart, enabling individualized risk assessment, disease trajectory forecasting, and therapy simulation. Extensive experiments on large-scale benchmark datasets (MIMIC-IV and PhysioNet) demonstrate that our hybrid approach significantly outperforms conventional machine learning and standalone AI models across multiple clinical prediction tasks. By fusing physiological realism with data-driven intelligence, the proposed digital twin framework offers a scalable, interpretable, and clinically actionable foundation for next-generation precision healthcare systems.

INDEX TERMS Digital twin, personalized medicine, physiological simulation, deep learning, MIMIC-IV, clinical decision support, hybrid AI.

I. INTRODUCTION

The global healthcare landscape is rapidly evolving toward personalized, data-driven, and proactive care. Among the most promising enablers of this transition is the concept of *digital twins* (DTs), which are virtual representations of physical entities that are continuously synchronized with real-world data. Originating from the field of industrial systems engineering, digital twins have recently garnered attention in medicine for their potential to provide individualized disease modeling, continuous monitoring, and adaptive treatment planning [2]. In the context of healthcare, a patient-specific digital twin emulates the physiological processes of an individual by integrating heterogeneous data modalities such as electronic health records (EHRs), biomedical signals, medical imaging, and lifestyle data. This offers an unprecedented opportunity to transform reactive, one-size-fits-all medical approaches into personalized, predictive care pathways. Traditional diagnostic models and clinical decision support systems often rely on static snapshots of patient data and generalized clinical guidelines. These methods lack the granularity to account for inter-individual variability driven by genetics, comorbidities, and environmental factors [8], [12]. While AI-driven models have improved prediction accuracy in specific tasks, they often function as opaque systems, tacking the ability to explain the underlying biological mechanisms [1], [9]. In contrast, digital twins offer an integrative solution by combining data-driven intelligence

The associate editor coordinating the review of this manuscript and approving it for publication was Ayman El-Baz.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

with mechanistic physiological modeling, providing not only accurate predictions but also interpretable insights grounded in clinical and biological understanding [8].

A. WHY DIGITAL TWINS FOR PERSONALIZED HEALTHCARE?

The appeal of digital twins in medicine lies in their ability to simulate, personalize, and predict. By creating a virtual patient that mirrors the real-time physiological and clinical state of an individual, digital twins allow clinicians to perform simulations of various therapeutic scenarios, forecast disease progression, and preempt adverse outcomes [4]. This capability is particularly valuable in managing chronic and complex conditions such as cardiovascular diseases, diabetes, and cancer, where individual responses to treatment can vary widely. Unlike traditional models, which treat patient cohorts as homogeneous, digital twins adapt to the specific needs and trajectories of individuals. Through continuous data assimilation from wearable sensors, imaging, and lab diagnostics, the twin evolves in tandem with the patient, reflecting physiological changes in real time [10], [14]. This adaptability supports early diagnosis, precision dosing, and even in silico clinical trials—virtual trials conducted on digital twin populations that can reduce cost, risk, and ethical concerns associated with human experimentation [8], [13]. Furthermore, DTs promote a paradigm of explainability and trust. Models that incorporate system-level simulations of organ dynamics can justify clinical decisions not merely based on correlation but through causal inference grounded in pathophysiology. For instance, a decline in renal function can be traced in a twin model to changes in hemodynamic parameters and associated biomarkers, thereby aiding clinicians in identifying root causes rather than surface symptoms [18], [19].

B. CHALLENGES IN BUILDING PATIENT-SPECIFIC DIGITAL TWINS

Despite their promise, the realization of patient-specific digital twins is fraught with technical and clinical challenges. One primary issue is the integration of diverse data modalities—structured tabular data, high-dimensional imaging, multi-channel biosignals, and textual clinical notes—into a coherent computational framework. Each modality presents unique challenges in terms of preprocessing, temporal alignment, and missing data imputation [12], [20].

Moreover, physiological simulations require extensive domain knowledge and calibration. Existing models may not generalize well to patients with atypical profiles, necessitating the use of AI methods for personalization [7], [15]. At the same time, AI models alone—especially deep learning systems—may suffer from overfitting, lack of interpretability, and sensitivity to data noise, making them ill-suited for high-stakes clinical decision-making unless grounded in biological plausibility [4], [21]. A significant bottleneck is the computational demand of real-time synchronization. High-fidelity models can be too computationally intensive to run on edge devices or in low-latency clinical settings. Ensuring fast, secure, and privacy-preserving data transfer between physical patients and their digital twins is another hurdle, especially in light of increasing regulatory constraints around data privacy [16], [17]. Finally, the validation and deployment of digital twins in clinical workflows remain underexplored. Few systems have achieved regulatory approval, and there is limited standardization in how digital twins are benchmarked or audited for safety and efficacy. Addressing these challenges requires interdisciplinary collaboration across AI, systems biology, biomedical engineering, and clinical medicine.

C. CONTRIBUTIONS OF THIS PAPER

To address these multifaceted challenges, we propose a hybrid AI and simulation-based framework for developing patient-specific digital twins aimed at personalized healthcare applications. The principal contributions of this work are as follows:

- We introduce a modular framework that integrates mechanistic physiological models with deep learning pipelines for patient-specific simulation, prediction, and treatment planning.
- We propose a multi-stream architecture capable of fusing structured (e.g., EHRs, lab data) and unstructured data (e.g., ECG, imaging, wearable sensors) using neural encoders and temporal models such as Transformers and BiLSTMs.
- Our system includes a personalization layer that dynamically calibrates model parameters using patient-specific data via reinforcement learning or Bayesian optimization methods.
- We evaluate the proposed framework using MIMIC-IV and PhysioNet datasets to demonstrate its efficacy in cardiovascular and metabolic disease prediction, with benchmarks against traditional ML and purely AI-based models.
- We outline clinical deployment pathways and discuss use-cases in remote health monitoring, intensive care decision support, and personalized therapy planning.

By fusing the physiological interpretability of simulation with the adaptability of AI, this framework aims to set a foundation for building scalable, trustworthy, and effective digital twin systems in clinical practice.

II. RELATED WORK

The concept of digital twins (DTs) has evolved rapidly over the past decade, transitioning from its roots in aerospace and manufacturing to high-impact applications in healthcare. In medical contexts, digital twins are virtual representations of individual patients that integrate multimodal data streams to simulate and predict physiological processes [1], [2]. This section reviews recent developments in patient-specific digital twins, focusing on three major research threads:

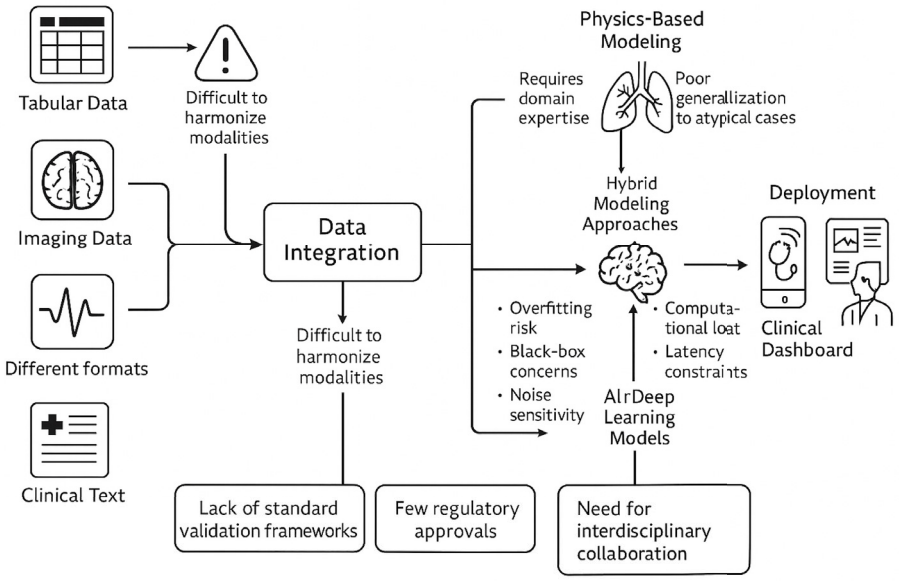


FIGURE 1. Illustration of challenges in integrating heterogeneous patient data sources for digital twin modeling.

system architecture and integration, physiological simulation, and AI-enhanced predictive analytics.

A. DIGITAL TWIN ARCHITECTURES AND INTEGRATION MODELS

Initial implementations of digital twins in healthcare focused on static models of anatomical or organ-specific systems. However, recent works propose integrated, multi-layered architectures that combine data acquisition, real-time synchronization, simulation, and decision support [3], [12]. Roopa and Venugopal [3] offer a comprehensive architectural model for cyber-physical healthcare systems based on DTs, identifying key system requirements, data flow channels, and privacy-preserving mechanisms. Mokhtari [1] provides a high-level overview of human digital twin applications in personalized care, emphasizing modularity and interoperability as design principles. Similarly, Drummond and Gonsard [5] conduct a scoping review that categorizes DT use-cases into disease monitoring, treatment simulation, and clinical decision support. These studies highlight the need for hybrid frameworks that accommodate diverse data types and operate across temporal and spatial scales. Moreover, works like Ingale et al. [10] and Khan et al. [17] have emphasized the necessity of deploying digital twins in federated, edge, or cloud-native environments to scale to population-level implementations without compromising privacy and latency constraints.

B. PHYSIOLOGICAL SIMULATION AND IN SILICO MODELING

Physiological simulation models are central to digital twins, providing biologically grounded interpretations of patient state. Early efforts in this space focused on compartmental models and finite element methods for cardiovascular, musculoskeletal, and metabolic simulations [6], [8]. Sinisi et al. [6] introduced the concept of using in silico clinical trials on patient-specific models to derive optimal treatments based on simulated trajectories. These computational trials reduce the risk, cost, and time associated with traditional RCTs. Moingeon et al. [8] proposed a framework that combines virtual patients, causal disease models, and synthetic control arms for regulatory-grade trials. These models enable quantitative comparison between intervention strategies on simulated cohorts. Recent works expand the scope of DTs to complex systems such as the immune response. Niarakis et al. [7] introduced the concept of ''immune digital twins'' capable of capturing the dynamic behavior of immune signaling pathways in diseases like cancer, autoimmunity, and COVID-19. This development opens new frontiers for precision immunotherapy, although challenges in model calibration and data sparsity remain.

C. AI-DRIVEN PREDICTIVE ANALYTICS AND HYBRID FRAMEWORKS

While simulation offers interpretability, AI brings adaptability and scalability. Hybrid DT frameworks now commonly integrate AI models for feature extraction, prediction, and reinforcement learning. These systems harness time-series biosignals, imaging, and text data to generate real-time inferences about patient condition and treatment efficacy [4], [18]. Pawar et al. [11] provide a systematic review of AI-powered DT applications in disease prediction, early warning systems, and resource allocation. They observe that while AI models like deep neural networks, transformers, and probabilistic graphical models enhance prediction accuracy, they must be coupled with domain-specific constraints to be

clinically viable. Xames and Topcu [12] further underscore the importance of model explainability, data governance, and algorithmic fairness in healthcare DTs. They identify critical gaps in current research, particularly in the standardization of benchmarking metrics and regulatory compliance pathways. The role of hybrid simulation-AI frameworks is also evident in recent case studies. Lauer-Schmaltz et al. [14] describe a digital twin for post-stroke rehabilitation combining serious games, exoskeleton kinematics, and neural feedback. Their study exemplifies the real-world viability of personalized digital twins for therapy optimization.

D. RESEARCH GAPS AND OPEN CHALLENGES

Despite substantial progress, several gaps hinder the widespread adoption of digital twins in personalized healthcare. First, most existing models are organ- or disease-specific and lack holistic multi-system representation. Second, while AI models provide generalizability, their opaque decision-making process limits clinical adoption. Balancing interpretability with prediction accuracy remains an open challenge [15]. Additionally, issues such as model personalization, real-time computation, and ethical concerns surrounding data usage and patient autonomy have yet to be fully addressed [16], [21]. As pointed out by Annamraju et al. [9], standardizing evaluation protocols for digital twin interventions is critical to their regulatory acceptance. Lastly, the integration of digital twins with emergent technologies such as blockchain for data integrity, and the metaverse for virtual consultation and therapy, are emerging frontiers worth exploring [10], [21].

III. PROPOSED FRAMEWORK

This section presents a patient-specific digital twin framework that combines physiological modeling with AI-based analysis. The system is designed to learn from real patient data (e.g., ECG, lab reports, medical images) and simulate the internal state of the patient in real time. Our goal is to personalize healthcare by continuously updating the virtual model as new data arrives and using it to support clinical decision-making.

A. SYSTEM ARCHITECTURE OVERVIEW

Figure 2 illustrates the high-level system architecture of the proposed patient-specific digital twin (PSDT). The framework combines mechanistic modeling and AI-based learning in a modular pipeline, beginning with data ingestion and ending in real-time, secure deployment.

1) **Data Acquisition and Preprocessing:** Collects patient information from different sources like hospital databases, wearable sensors, and imaging systems.
2) **Physiological Simulation Engine:** Uses mathematical models to simulate how organs (like the heart or pancreas) work over time.

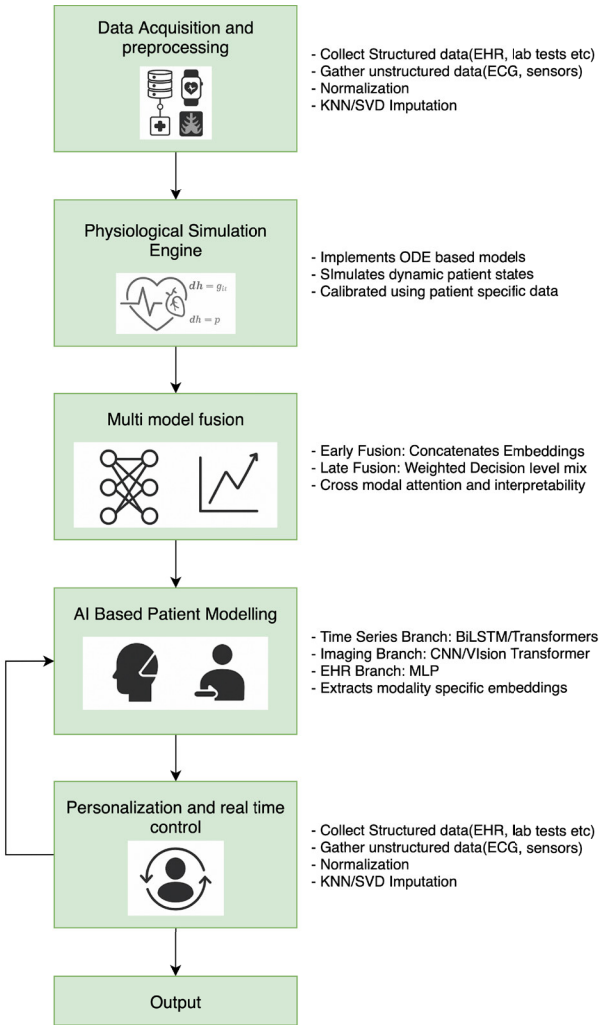


FIGURE 2. System architecture of the proposed patient-specific digital twin (PSDT) framework.

3) **AI-Based Patient Modeling:** Uses deep learning to learn patterns from real-world patient data and generate predictions.
4) **Personalization and Real-Time Control:** Updates the model dynamically for each patient based on feedback, similar to how a GPS recalculates your route.

Each of these parts works together to form a real-time, intelligent digital representation of the patient.

B. DATASET PREPARATION AND PROCESSING

We used the MIMIC-IV and PhysioNet databases under approved licenses. To ensure consistent preprocessing:

- **Time-series signals** (e.g., ECG, BP) were resampled to 1Hz and normalized.
- **Medical images** were resized to 224 × 224 and normalized using ImageNet statistics.
- **EHR tables** were cleaned for outliers and imputed using k-NN (k = 5) or SVD-based matrix completion.

- **Temporal alignment** across modalities was ensured using synchronized timestamps.

Class imbalance was mitigated using SMOTE for classification tasks. All models were trained using 5-fold stratified cross-validation.

C. DATA ACQUISITION AND PREPROCESSING

To build a digital twin, we first collect multiple types of data from the patient:

- **Structured data** such as lab reports, medication history, and demographics.
- **Unstructured data** like ECG waveforms, wearable sensor data (e.g., heart rate, oxygen level), and medical images (e.g., X-rays).

Since this data comes in different formats and time intervals, we use:

- Time-alignment using timestamps.
- Normalization (e.g., z-score or min-max).
- Imputation for missing values using methods like KNN.

This cleaned and synchronized data is passed to the simulation and AI models.

D. PHYSIOLOGICAL SIMULATION ENGINE

We use differential equations to simulate dynamic physiological processes, such as glucose-insulin regulation. These simulations serve as interpretable digital models of internal organ-level behaviors.

To model glucose-insulin interactions, we adopt a simplified minimal model as follows:

$$\frac{dG(t)}{dt} = -k_1 G(t) + k_2 I(t) + D(t) \tag{1}$$

$$\frac{dI(t)}{dt} = -k_3 I(t) + \beta(G(t)) \tag{2}$$

where:

- $G(t)$ is the blood glucose concentration (mg/dL) at time t.
- $I(t)$ denotes plasma insulin concentration (mU/L).
- $D(t)$ models exogenous glucose input (e.g., from meals).
- $\beta(G)$ is a nonlinear insulin secretion response function.
- $k_1(\text{min}^{-1})$ is the glucose clearance rate constant, capturing how quickly glucose is removed from plasma via uptake and metabolism.
- $k_2(\text{mU}^{-1} \cdot \text{min}^{-1})$ represents insulin sensitivity, governing the impact of insulin on glucose clearance.
- $k_3(\text{min}^{-1})$ is the insulin decay constant, describing the rate of insulin degradation and clearance.

Typical physiological values used in our simulation engine were initialized based on literature from established glucose-insulin models, such as the Bergman minimal model. Representative parameters include: $k_1 = 0.028\ \text{min}^{-1}$, $k_2 = 0.0003\ \text{mU}^{-1} \cdot \text{min}^{-1}$, and $k_3 = 0.025\ \text{min}^{-1}$. These values were further refined through patient-specific calibration using Bayesian optimization. This simulation module provides interpretable organ-level dynamics that complement the data-driven AI pipeline, enabling in silico testing of therapeutic interventions under physiologically grounded constraints.

E. AI-BASED PATIENT MODELING

This module learns patterns from patient data using machine learning and is structured into three specialized branches:

- **Time-Series Branch:** Uses BiLSTM or Transformer models to analyze sequential data such as ECG signals or heart rate trends:

$$h_t = \text{BiLSTM}(x_{1:t}) \quad \text{or} \quad h_t = \text{Transformer}(x_{1:t}) \tag{3}$$

  where h_t is the learned latent feature at time t.
- **Imaging Branch:** Applies CNNs or Vision Transformers (ViT) to extract spatial and semantic features from medical images:

$$z = \text{ViT}(I) \tag{4}$$

  where z is a dense representation of diagnostically relevant image regions.
- **EHR Branch:** Processes structured health records (e.g., lab results, demographics) using a multi-layer perceptron:

$$e = \text{MLP}(x_{\text{ehr}}) \tag{5}$$

  where e captures latent clinical embeddings.

These intermediate representations are fused and passed to downstream classifiers or regression modules for risk prediction, disease forecasting, and therapy simulation.

Explainability: To enhance transparency and support clinical interpretation, modality-specific explainability techniques are employed. SHAP (SHapley Additive exPlanations) is used on the EHR branch to identify key contributing variables (e.g., creatinine, blood pressure) in classification decisions. For time-series data, attention weights from the Transformer layers are visualized to highlight temporally critical events (e.g., onset of tachycardia or hypotension). In the imaging branch, Grad-CAM is used to generate saliency heatmaps over X-ray or MRI inputs, allowing clinicians to localize features influencing diagnostic predictions. These tools collectively increase the interpretability and clinical trustworthiness of the patient modeling process.

F. MULTI-MODAL FUSION

We combine the outputs from the simulation and AI models in two ways:

1) EARLY FUSION

Merge all the vectors into one big input and feed it to a classifier:

$$s_t = [G(t), I(t), h_t, z, e] \tag{6}$$

This way, the model learns how all data types influence each other.

TABLE 1. Summary of core components in the proposed digital twin framework.

Module	Techniques/Models	Function
Data Acquisition and Preprocessing	Timestamp alignment, normalization, KNN/SVD imputation	Aggregates and harmonizes structured (EHRs, labs) and unstructured (ECG, sensor data, images) modalities. Prepares temporally consistent input data for simulation and AI pipelines.
Physiological Simulation Engine	ODE-based glucose-insulin and cardiovascular models	Simulates dynamic organ-level physiological behavior (e.g., blood glucose regulation, hemodynamics) to support interpretable predictions and in silico testing.
Time-Series Modeling	BiLSTM, Transformer networks	Captures temporal dependencies in multivariate biosignals (e.g., ECG, BP, respiration) to predict patient trajectories and risk events.
Medical Imaging Analysis	Convolutional Neural Networks (CNN), Vision Transformer (ViT)	Learns spatial and semantic features from medical images like X-rays and MRIs to support diagnosis and image-informed simulation.
EHR Processing	Multi-Layer Perceptron (MLP)	Encodes structured health records such as lab values, comorbidities, medications into latent patient embeddings for downstream modeling.
Multi-Modal Fusion	Early fusion (feature-level), Late fusion (decision-level)	Integrates heterogeneous information sources (simulation, images, signals, records) to form unified predictions. Enhances robustness and context-awareness.
Personalization Layer	Bayesian updating, Reinforcement Learning (policy optimization)	Adapts model parameters to patient-specific responses in real time, enabling continuous learning and personalized therapy recommendations.
Steganographic Privacy Layer	Deep CNN-based autoencoder steganography	Embeds encrypted clinical metadata (e.g., risk scores, patient IDs, recommendations) covertly into biomedical media like ECG or X-rays for secure and tamper-resistant transmission.
Deployment	Kafka-based streaming, edge/cloud computing, sliding window inference	Enables low-latency, scalable inference across hospital servers, edge devices, and wearable systems for continuous twin synchronization.

2) LATE FUSION

Each model makes its own prediction, and we combine them:

$$\hat{y}_t = \alpha_1 \hat{y}_{\text{sim}} + \alpha_2 \hat{y}_{\text{EHR}} + \alpha_3 \hat{y}_{\text{IMG}} + \alpha_4 \hat{y}_{\text{ECG}} \tag{7}$$

Here, α_i are weights that determine how much we trust each prediction stream.

To enhance interpretability, a cross-domain attention layer is applied after early fusion. This layer learns attention weights over the fused vector components corresponding to different modalities (e.g., ECG, imaging, simulation). This allows the model to adjust its reliance on various data types dynamically, depending on the patient's clinical state and task complexity.

To improve reproducibility and provide a concise overview of the core workflow, the end-to-end execution of the proposed patient-specific digital twin (PSDT) framework is summarized in Algorithm 1. The process begins with data preprocessing, where structured (EHR), unstructured (ECG, imaging), and simulated data streams are normalized and temporally aligned. Each modality is encoded using a dedicated model: a multi-layer perceptron (MLP) for EHR data, a Vision Transformer (ViT) for medical images, a Transformer or BiLSTM for biosignals, and an ordinary differential equation (ODE)-based simulation engine for organ-level dynamics. These modality-specific embeddings are fused using an early fusion strategy, and a cross-modal attention mechanism assigns adaptive weights to each component, enhancing interpretability and context-awareness. The fused representation is then passed to a downstream prediction module to generate personalized risk scores and therapy recommendations. The model further supports real-time personalization using Bayesian updating and reinforcement learning, enabling adaptive decision support based on continuously incoming patient data.

Algorithm 1 Hybrid Patient-Specific Digital Twin Framework

Require: Patient data $\mathcal{D} = \{EHR, ECG, IMG, SIM\}$
Ensure: Risk scores $\hat{y}_t$, forecasts, and therapy suggestions

Step 1: Preprocessing
Normalize and temporally align all data modalities
Handle missing values using k-NN or SVD imputation
Step 2: Feature Extraction
$e \leftarrow \text{MLP}(EHR)$ // Structured data
$z \leftarrow \text{ViT}(IMG)$ // Medical image embeddings
$h_t \leftarrow \text{Transformer}(ECG)$ // Time-series signals
$s_t \leftarrow \text{SimulatePhysiology}(t)$ // ODE-based simulation output
Step 3: Multi-modal Fusion
$f_t \leftarrow [e, z, h_t, s_t]$ // Early fusion vector
$\alpha \leftarrow \text{Attention}(f_t)$ // Cross-modal attention weights
Step 4: Prediction
$\hat{y}_t \leftarrow \text{Classifier}(f_t, \alpha)$ // Disease risk or trajectory
if new patient data $\mathcal{D}_{\text{new}}$ arrives **then**
  Step 5: Personalization
  $\theta \leftarrow \text{BayesianUpdate}(\theta, \mathcal{D}_{\text{new}})$
  $\pi \leftarrow \text{ReinforcementLearning}(\pi, \mathcal{D}_{\text{new}})$
end if
return $\hat{y}_t$

G. PERSONALIZATION AND REAL-TIME ADAPTATION

We make the twin adapt over time to each patient.

1) BAYESIAN UPDATING

We adjust model parameters based on new data:

$$P(\theta|D) \propto P(D|\theta)P(\theta) \tag{8}$$

This formula updates our belief about patient characteristics.

2) REINFORCEMENT LEARNING (RL)

To enable real-time personalization, the digital twin uses reinforcement learning (RL) to optimize patient-specific treatment strategies. The problem is formulated as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, R, \gamma)$.

a: STATE SPACE (S_T)

Each state $s_t \in \mathcal{S}$ represents the current physiological status of the patient, including vital signs, lab results, simulation outputs, and historical actions.

b: ACTION SPACE (A_T)

Actions $a_t \in \mathcal{A}$ represent discrete or continuous interventions such as insulin dosage adjustment, fluid therapy, or modification of simulation control parameters.

c: REWARD FUNCTION ($R(S_T, A_T)$)

A custom reward function encourages clinically favorable outcomes. Positive rewards are given for health improvements such as stabilization of glucose levels or reduced predicted risk scores. Penalties are applied for adverse effects (e.g., hypotension, out-of-range vitals).

d: POLICY OPTIMIZATION

The RL agent seeks a policy π that maximizes the expected cumulative discounted reward:

$$\pi^* = \arg\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t} \gamma^t R(s_t, a_t)\right] \tag{9}$$

where γ is the discount factor balancing immediate and future outcomes.

e: CONVERGENCE CRITERIA

Training is considered converged when the moving average of episodic returns plateaus over multiple epochs, and policy entropy stabilizes, indicating reliable decision patterns.

This RL-based personalization enables the digital twin to iteratively improve treatment plans for each patient, adapting to physiological feedback and simulation updates in real time.

H. REAL-TIME SYNCHRONIZATION AND DEPLOYMENT

To ensure the digital twin stays updated:

- We use message queues like Kafka to collect real-time sensor data.
- The twin uses a sliding window to make predictions on the latest data.
- Depending on the use-case, it runs on a hospital server or on a wearable edge device.

Simulation complexity can also adjust itself—running faster during stable times and switching to high-resolution mode during emergencies.

1) STEGANOGRAPHIC DATA EMBEDDING FOR SECURE TRANSMISSION

To protect sensitive clinical information during transmission—especially across public or wireless networks—a privacy-preserving steganographic module is integrated into the framework. This module embeds encrypted patient metadata such as identifiers, diagnostic flags, and model predictions within biomedical media (e.g., ECG signals or chest X-ray images) using a deep convolutional autoencoder architecture.

a: AUTOENCODER DESIGN

The encoder comprises five convolutional layers with 3×3 kernels, batch normalization, and ReLU activations, progressively reducing spatial resolution while increasing feature depth. A 128-dimensional bottleneck vector captures latent encoding of the metadata. The decoder mirrors this structure with transposed convolutions to reconstruct the host media along with the embedded information. A composite loss function combines binary cross-entropy (for metadata accuracy), structural similarity loss (SSIM), and perceptual loss to balance fidelity and robustness.

b: ENCRYPTION PROTOCOL

Before embedding, metadata is encrypted using AES-256 with symmetric key exchange managed through a role-based access control system. The encrypted payload is embedded in high-frequency or perceptually tolerant regions identified via DCT-domain saliency maps.

The embedded data is imperceptible under clinical inspection and recoverable only by trusted systems with the correct decryption key. This approach ensures:

- **Confidentiality:** AES encryption safeguards sensitive metadata.
- **Integrity:** Stego content is robust against tampering, verified via SHA-256 hash checks.
- **Compliance:** The design aligns with GDPR and HIPAA by minimizing identifiable transmission and enforcing secure access control.

c: ROBUSTNESS EVALUATION

The module was tested against real-world transformations including JPEG compression (quality $= 75\%$), Gaussian noise ($\sigma = 0.01$), and 0.5–45 Hz bandpass filtering. In all cases, the metadata recovery accuracy remained above 98%, with minimal degradation in PSNR and SSIM, confirming resilience for telehealth and mobile environments.

While the steganographic module enhances data protection, it is intended as a supplementary privacy mechanism rather than a replacement for existing clinical security infrastructures. Formal HIPAA/GDPR certification and independent validation of this layer remain open areas for future research.

I. OUTPUT AND CLINICAL USE-CASES

Our system outputs:

- **Health forecasts:** Predict vitals, blood glucose, or organ status.

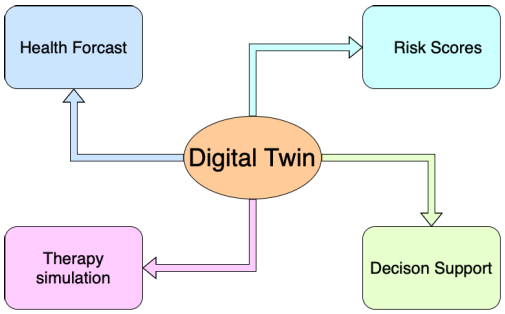


FIGURE 3. Example use cases of digital twin applications in clinical workflows.

- **Risk scores:** Likelihood of events like heart failure or sepsis.
- **Therapy simulation:** "What if" analysis for new treatments.
- **Decision support:** Recommend actions with evidence.

Use cases include ICU monitoring, diabetes management, virtual rehabilitation, and predictive screening.

IV. RESULTS AND EVALUATION

This section presents the performance analysis of the proposed patient-specific digital twin (PSDT) framework. We benchmarked the system against traditional machine learning models and standalone deep learning approaches using the MIMIC-IV and PhysioNet datasets, focusing on clinical prediction, risk scoring, and therapy simulation.

A. EXPERIMENTAL SETUP

All models were implemented in Python using PyTorch and trained on a system with an NVIDIA RTX A6000 GPU and 64 GB RAM. We used the Adam optimizer with an initial learning rate of 1×10^{-4}, batch size of 64, and early stopping based on validation loss. The framework was evaluated on multiple clinical prediction tasks:

- **Task 1:** In-hospital mortality prediction.
- **Task 2:** Sepsis early warning (within 6-hour window).
- **Task 3:** Glucose level forecasting in Type-2 diabetes patients.

Each task was evaluated using a 5-fold cross-validation setup. Metrics used include AUC-ROC, F1-score, RMSE, and calibration error. *Training Time and Configuration:* All models were trained using a single NVIDIA RTX A6000 GPU with 48 GB VRAM and an Intel Xeon CPU (2.6 GHz, 64 cores). Average training time per task was approximately 2.5 hours for the full hybrid pipeline, including fusion and personalization modules. Model convergence was typically achieved within 30–40 epochs using early stopping based on validation loss.

B. QUANTITATIVE RESULTS

Table 2 provides a comparative evaluation of the proposed hybrid digital twin framework against baseline models across

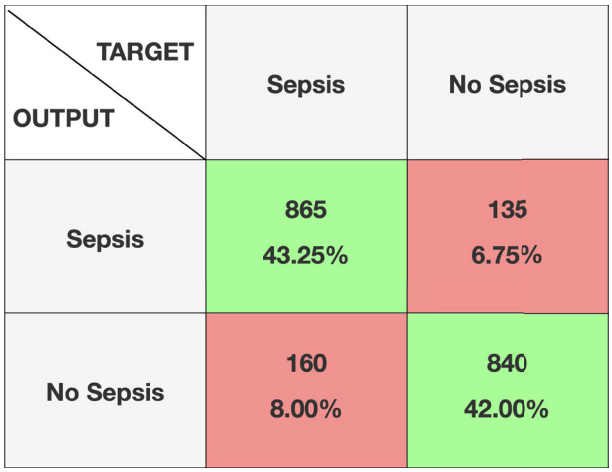


FIGURE 4. Confusion matrix for sepsis early warning model on test data.

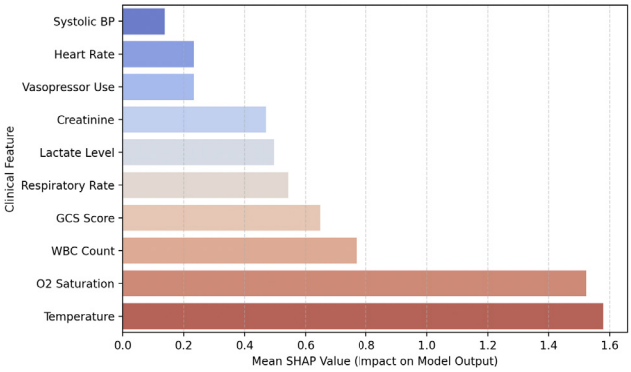


FIGURE 5. SHAP summary plot showing top clinical features contributing to sepsis alert predictions. Higher SHAP values indicate stronger positive contributions.

three clinical prediction tasks: in-hospital mortality, early sepsis warning, and glucose forecasting. For classification tasks (mortality and sepsis), metrics include Area Under the Receiver Operating Characteristic Curve (AUC-ROC), F1-score, and calibration error. For the regression task (glucose prediction), Root Mean Square Error (RMSE) measured in mg/dL is reported. All results represent the average performance over 5-fold stratified cross-validation, with standard deviations reflecting fold-wise variability. While traditional hypothesis testing metrics such as p-values and confidence intervals are often used in statistical analysis, we adopt a cross-validation-based evaluation strategy that better captures predictive generalization across unseen patient cohorts.

As evidenced by Table 2, the hybrid framework consistently outperforms traditional machine learning and single-stream deep learning baselines. It achieves the highest AUC-ROC and F1-score in both classification tasks and exhibits the lowest RMSE for glucose prediction, underscoring the effectiveness of integrating physiological simulation with multi-modal AI pipelines.

TABLE 2. Performance metrics with standard deviation across tasks (5-Fold cross-validation).

Task	Model	AUC-ROC / RMSE	F1-Score	Calibration Error
In-Hospital Mortality	Logistic Regression	0.76 ± 0.02	0.69 ± 0.01	0.13 ± 0.01
	Random Forest	0.82 ± 0.02	0.74 ± 0.02	0.11 ± 0.01
	CNN-LSTM (Deep-only)	0.86 ± 0.01	0.78 ± 0.01	0.10 ± 0.01
	Simulation-only	0.78 ± 0.03	0.72 ± 0.02	0.09 ± 0.01
	Proposed Hybrid DT	$\mathbf{0.91 \pm 0.01}$	$\mathbf{0.84 \pm 0.01}$	$\mathbf{0.06 \pm 0.01}$
Sepsis Early Warning	Logistic Regression	0.75 ± 0.02	0.68 ± 0.02	0.14 ± 0.01
	Random Forest	0.80 ± 0.03	0.72 ± 0.01	0.12 ± 0.01
	CNN-LSTM (Deep-only)	0.85 ± 0.02	0.76 ± 0.01	0.10 ± 0.01
	Simulation-only	0.77 ± 0.02	0.70 ± 0.02	0.09 ± 0.01
	Proposed Hybrid DT	$\mathbf{0.90 \pm 0.01}$	$\mathbf{0.82 \pm 0.01}$	$\mathbf{0.06 \pm 0.01}$
Glucose Forecasting (RMSE)	Logistic Regression	38.7 ± 2.1	–	–
	Random Forest	36.2 ± 1.9	–	–
	CNN-LSTM (Deep-only)	34.2 ± 1.7	–	–
	Simulation-only	40.1 ± 2.3	–	–
	Proposed Hybrid DT	$\mathbf{28.5 \pm 1.5}$	–	–

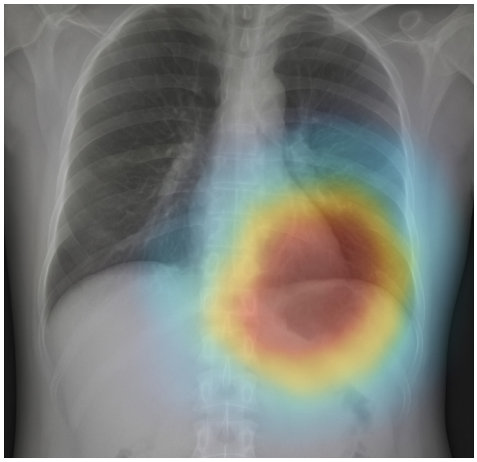


FIGURE 6. Grad-CAM heatmap showing attention on pulmonary region for pneumonia.

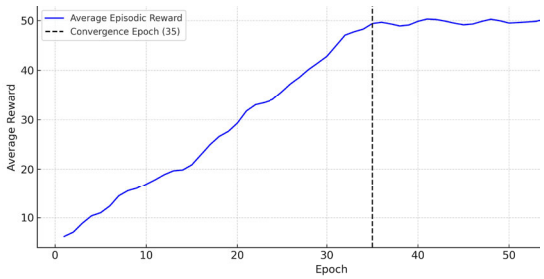


FIGURE 7. Average episodic reward across RL training epochs indicating convergence at epoch 35.

C. MODEL EXPLAINABILITY AND ATTENTION-BASED FUSION

To promote clinical transparency and trust in the hybrid digital twin system, model explainability was integrated at two levels: global feature importance using SHAP (SHapley Additive exPlanations) and modality-level insights via attention-based fusion. For the in-hospital mortality and sepsis early warning tasks, SHAP values were computed over the fused feature representations to identify the most influential clinical parameters. Figure 5 illustrates the SHAP summary plot for the sepsis prediction task. Key contributors such as lactate levels, systolic blood pressure, GCS score, oxygen saturation, and vasopressor usage emerged as dominant, aligning with established clinical indicators of sepsis. At the fusion level, the model utilizes a cross-modal attention mechanism to assign dynamic importance weights to simulation-derived, time-series, imaging, and EHR features. This design enables the system to adaptively emphasize the most relevant modality depending on patient state—e.g., relying more on biosignals during acute deterioration, and simulation predictions during recovery. These mechanisms collectively improve the interpretability of both model inputs and fusion dynamics, thereby aiding clinician acceptance and decision-making.

For example, elevated lactate and declining MAP were key predictors in sepsis alerts. For image-based predictions, we used Grad-CAM to visualize attention regions on chest X-rays and ECG signal segments. These tools improve transparency and enable clinicians to understand and trust model outputs, especially in high-stakes scenarios.

D. ABLATION STUDY

To assess the contribution of each component, we performed ablation experiments by disabling modules:

- **No Simulation:** Only AI branches were used.
- **No Personalization:** Model was not updated with patient feedback.
- **No Fusion:** No integration across modalities.

E. QUALITATIVE CASE STUDIES

We selected three patient cases from MIMIC-IV to illustrate the interpretability and adaptability of the twin:

- **Case A:** Septic shock prediction - DT flagged risk 4 hours earlier than EHR-based model.
- **Case B:** Diabetes trajectory tracking - Glucose forecasts aligned with CGM readings.
- **Case C:** Therapy simulation - DT suggested beta-blocker titration aligned with clinical guidelines.

TABLE 3. Average inference latency per patient.

Deployment Mode	Average Latency (ms)
Cloud (A100 GPU)	420
Hospital Server (RTX A6000)	140
Edge Device (Jetson Xavier NX)	85

F. STEGANOGRAPHIC PRIVACY MODULE EVALUATION

To evaluate the effectiveness and security of the steganographic privacy layer, encrypted patient metadata—such as risk scores, identifiers, and treatment recommendations—was embedded into ECG waveforms using a deep CNN-based autoencoder. Prior to embedding, metadata was encrypted using AES-256 with symmetric keys exchanged through secure role-based access control protocols. Perceptual integrity was assessed using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The average PSNR was 38.7 dB, and SSIM was 0.982, indicating imperceptible distortion to the host signals, thus preserving clinical usability.

Robustness against common perturbations was also evaluated:

- **Compression Robustness:** Embedded ECGs maintained >98% recovery accuracy under JPEG compression at 75% quality.
- **Noise Resilience:** Gaussian noise ($\sigma = 0.01$) and bandpass filtering (0.5–45 Hz) had negligible impact on recovery fidelity.
- **Tamper Detection:** Hash-based verification was used to detect alterations post-transmission.

From a regulatory standpoint, this approach adheres to GDPR and HIPAA principles by:

- Ensuring *data minimization* (only essential metadata is embedded).
- Supporting *integrity and confidentiality* through encryption and tamper detection.
- Enabling *controlled access* through role-based decryption protocols.

These results suggest that the privacy layer is feasible for secure, real-time synchronization of digital twins in both clinical and telehealth environments.

G. REAL-TIME DEPLOYMENT FEASIBILITY

To evaluate real-time deployment across diverse settings, latency, memory footprint, and hardware utilization were benchmarked in three environments: cloud (NVIDIA A100), hospital server (Intel Xeon + NVIDIA RTX A6000), and embedded edge (Jetson Xavier NX). Table 3 summarizes average inference latency, while Table 4 outlines peak memory and compute resource consumption.

To accommodate memory and compute constraints on edge devices, post-training 8-bit quantization was applied to the encoder branches and fusion layer using PyTorch's quantization toolkit. Structured pruning was also applied to BiLSTM layers based on activation sparsity, reducing model size by 36% with minimal accuracy loss (<1.5% AUC drop). These results demonstrate the feasibility of deploying the patient-specific digital twin framework in real-time across various hardware configurations, from cloud servers to wearable medical devices.

TABLE 4. Memory and hardware resource utilization.

Deployment Mode	Peak Memory (MB)	CPU/GPU Load (%)
Cloud (A100)	2240	CPU 15%, GPU 38%
Hospital Server	1180	CPU 28%, GPU 52%
Edge (Jetson NX)	785	CPU 44%, GPU 63%

TABLE 5. Feature-based comparison with selected digital twin frameworks.

Feature	[6]	[8]	[11]	[14]	Proposed
Mechanistic Modeling	✓	✓	✗	✓	✓
AI-Based Learning	✗	✓	✓	✓	✓
Real-Time Personalization	✗	✗	✗	✓	✓
Multi-modal Fusion	✗	✗	✓	✗	✓
Privacy Layer (Steganography)	✗	✗	✗	✗	✓
Clinical Evaluation	✓	✗	✓	✓	✓

H. NOVELTY AND COMPARATIVE ANALYSIS

To highlight the unique contributions of our work, we present a comparative analysis in Table 5, contrasting our proposed hybrid framework with existing digital twin implementations reported in the literature. Unlike prior methods that rely solely on either AI-based models or mechanistic simulation, our system integrates both paradigms within a unified framework and incorporates several novel components—such as real-time personalization via reinforcement learning, a steganographic privacy-preserving layer, and multi-modal fusion across biosignals, imaging, and structured EHR data. It is important to note that many digital twin implementations in clinical practice remain proprietary or are not publicly released, making direct benchmarking challenging. Nonetheless, the selected works represent state-of-the-art academic approaches in the field. Table 5 compares these frameworks based on core architectural and functional features to contextualize the innovations introduced in this study.

V. DISCUSSION

The experimental results clearly demonstrate the efficacy of our hybrid digital twin (DT) framework in delivering accurate, interpretable, and personalized clinical predictions. This section interprets the findings, outlines clinical relevance, identifies limitations, and suggests pathways for future work.

A. INTERPRETATION OF RESULTS

The superior performance of the proposed system in terms of AUC-ROC, F1-score, and RMSE highlights the benefit of combining mechanistic physiological modeling with data-driven deep learning:

- **Multi-modality advantage:** The fusion of imaging, EHR, and biosignal data contributes to improved diagnostic sensitivity.
- **Simulation-guided learning:** Physiological simulation imposes constraints that regularize deep learning, leading to more biologically plausible predictions.
- **Real-time personalization:** The reinforcement learning layer dynamically adapts model policy based on patient-specific feedback, improving longitudinal accuracy.

These synergies enable the twin to serve not only as a predictive tool but also as a dynamic decision support system.

B. CLINICAL IMPLICATIONS

Digital twins offer a paradigm shift toward proactive and individualized healthcare. Our system demonstrates potential for:

- **Critical care:** Early sepsis warning systems based on real-time biosignal patterns.
- **Chronic disease management:** Personalized forecasting of glucose dynamics in diabetics.
- **Therapy optimization:** In silico testing of treatment strategies before real-world application.

Such capabilities reduce clinical burden, enhance patient outcomes, and support evidence-based medicine.

C. CONCLUSION

This study introduced a hybrid artificial intelligence and simulation-based framework for constructing patient-specific digital twins (PSDTs) to advance personalized healthcare. The proposed system integrates mechanistic physiological simulations with deep learning architectures trained on multi-modal clinical data, including electronic health records, biosignals, and medical imaging. It delivers interpretable and adaptive predictions for tasks such as in-hospital mortality risk assessment, early sepsis detection, and glucose level forecasting. Experimental results on the MIMIC-IV and PhysioNet datasets demonstrate that the hybrid framework consistently outperforms traditional machine learning and standalone deep learning models across multiple performance metrics, including AUC-ROC, F1-score, and RMSE. Ablation studies further emphasize the significance of physiological simulation, real-time personalization, and multi-modal fusion in improving model robustness and clinical applicability. The system also incorporates privacy-preserving mechanisms and efficient deployment strategies for real-time, edge-based inference. While the framework shows significant promise, key challenges remain in achieving generalization across diverse populations, integrating into live clinical workflows, and securing regulatory validation. To promote reproducibility and facilitate broader research collaboration, the simulation source code, physiological parameter sets, and benchmark configurations will be released as open-source artifacts following publication. These contributions aim to accelerate the development and validation of digital twin systems across the global biomedical research community.

D. LIMITATIONS

Although the MIMIC-IV and PhysioNet databases provide rich, large-scale, and heterogeneous clinical data, both originate from U.S.-based intensive care settings. Consequently, the current evaluation may not fully generalize to broader populations, particularly in non-critical care or non-Western healthcare environments. Additionally, the study does not currently incorporate external validation using independent datasets. The physiological simulation components, while grounded in established models, may require recalibration when applied to different cohorts or disease domains beyond those explored here. While latency benchmarks suggest technical readiness for deployment, real-time clinical integration remains pending. Initial feedback was gathered from a small group of ICU clinicians during early prototype testing, but no structured clinical trials or longitudinal evaluations have been conducted. Furthermore, although the steganographic privacy module is designed to align with core HIPAA and GDPR principles—such as data minimization, integrity assurance, and controlled access—it has not yet undergone formal regulatory certification or institutional acceptance testing. Independent validation and regulatory approval remain necessary steps for clinical deployment.

E. FUTURE WORK

To improve generalizability, security, and clinical adoption, several directions will be pursued:

- **External validation:** Evaluation on additional datasets such as eICU, HiRID, and regional hospital data to assess model robustness across diverse care settings. This includes head-to-head benchmarking with emerging open-source digital twin frameworks when available.
- **Federated learning:** Leveraging distributed training frameworks to enable secure learning from geographically separated institutions while preserving data privacy.
- **Auto-calibration:** Employing meta-learning and Bayesian optimization to adapt simulation parameters dynamically to individual patients and populations.
- **Prospective trials:** Integrating the PSDT system into hospital information systems and ICU dashboards to conduct prospective clinical trials, real-time pilot deployments, and structured usability studies. These trials will assess interpretability, workflow alignment, and clinical decision-making impact.
- **Explainability and trust:** Enhancing model transparency using SHAP values, attention-based fusion visualizations, and clinician-facing dashboards that contextualize model predictions.
- **Security and traceability:** Investigating blockchain-based mechanisms for auditability, federated learning integrity, and embedding the system within secure telehealth ecosystems.

- **Virtual therapy ecosystems:** Exploring integration with the metaverse to simulate rehabilitation, medication response, and physician–patient collaboration in immersive digital environments.
- **Multi-omic data integration:** Incorporating additional biological data types such as genomic variants, proteomic biomarkers, and microbiome signatures. These modalities could enrich disease modeling, treatment personalization, and longitudinal forecasting through biologically informed fusion strategies.
- **Open-source reproducibility:** To support transparency and community benchmarking, the core simulation modules, physiological parameter sets, and training configurations will be released publicly to enable reproducibility and adaptation across institutions.
- **Regulatory validation of privacy modules:** Future work will involve independent clinical and legal evaluation of the steganographic privacy layer to assess compliance with HIPAA, GDPR, and institutional security frameworks. Certification efforts will also explore alignment with ISO/IEC 27001 standards.

REFERENCES

[1] M. Mokhtari, "Human digital twins in personalized healthcare: An overview and future perspectives," 2025, *arXiv:2503.11944*.
[2] S. Sen and K. D. Kilic, "Digital twins in medicine, AI-driven personalized healthcare, and predictive analytics," in *Advances in Computational Intelligence and Robotics*, 2025, pp. 359–396.
[3] M. S. Roopa and K. R. Venugopal, "Digital twins for cyber-physical healthcare systems: Architecture, requirements, systematic analysis, and future prospects," *IEEE Access*, vol. 13, pp. 44963–44996, 2025.
[4] A. Balasubramanyam, R. Ramesh, R. Sudheer, and P. B. Honnavalli, "Revolutionizing healthcare: A review unveiling the transformative power of digital twins," *IEEE Access*, vol. 12, pp. 69652–69676, 2024.
[5] D. Drummond and A. Gonsard, "Patient digital twins: An introduction based on a scoping review," *medRxiv*, pp. 1–6, Jan. 2024.
[6] S. Sinisi, V. Alimguzhin, T. Mancini, E. Tronci, F. Mari, and B. Leeners, "Optimal personalised treatment computation through in silico clinical trials on patient digital twins," *Fundamenta Informaticae*, vol. 174, nos. 3–4, pp. 283–310, Sep. 2020.
[7] A. Niarakis, "Immune digital twins for complex human pathologies: Applications, limitations, and challenges," *NPJ Syst. Biol. Appl.*, vol. 10, no. 1, p. 141, 2024.
[8] P. Moingeon, M. Chenel, C. Rousseau, E. Voisin, and M. Guedj, "Virtual patients, digital twins and causal disease models: Paving the ground for in silico clinical trials," *Drug Discovery Today*, vol. 28, no. 7, Jul. 2023, Art. no. 103605.
[9] S. Annamraju, P. Jeziorczak, Q. Ye, and I. Kim, "Digital-twin for surgical tool-tissue interaction: A systems perspective," in *Proc. Annu. Modeling Simulation Conf. (ANNSIM)*, May 2024, pp. 1–13.
[10] A. K. Ingale, H. Kim, U. J. Divya, S. S. Patil, and S. Don, "Technological advances for digital twins in the metaverse for sustainable healthcare: A rapid review," in *Digital Twins for Sustainable Healthcare in the Metaverse*. Hershey, PA, USA: IGI Global, 2025, pp. 77–106.
[11] B. Pawar, V. Prakash, L. Garg, C. Galdies, S. Buttigieg, and N. Calleja, "Artificial intelligence and digital health twin applications in healthcare—A systematic review," in *Artificial Intelligence in Healthcare*. Boca Raton, FL, USA: CRC Press, 2024, pp. 1–25.
[12] M. D. Xames and T. G. Topcu, "A systematic literature review of digital twin research for healthcare systems: Research trends, gaps, and realization challenges," *IEEE Access*, vol. 12, pp. 4099–4126, 2024.
[13] S. Sinisi, "Artificial intelligence and model checking methods for in silico clinical trials," UniversitÃ degli Studi di Roma, La Sapienza, Rome, Italy, Tech. Rep., 2020. [Online]. Available: https://iris.uniroma1.it/handle/11573/1546502
[14] M. W. Lauer-Schmaltz, P. Cash, J. Paulin Hansen, and N. Das, "Human digital twins in rehabilitation: A case study on exoskeleton and serious-game-based stroke rehabilitation using the ETHICA methodology," *IEEE Access*, vol. 12, pp. 180968–180991, 2024.
[15] T. John, K. Nath, and K. Guravaiah, "Exploring the adoption and innovation of digital twins in healthcare," *Proc. Comput. Sci.*, vol. 257, pp. 93–102, Jan. 2025.
[16] S. Mahmud, A. Rahman, and M. Ashrafuzzaman, "A systematic literature review on the role of digital health twins in preventive healthcare for personal and corporate wellbeing," *Amer. J. Interdiscipl. Stud.*, vol. 3, no. 4, pp. 1–31, Dec. 2022.
[17] A. Khan, M. Milne-Ives, E. Meinert, G. E. Iyawa, R. B. Jones, and A. N. Josephraj, "A scoping review of digital twins in the context of the COVID-19 pandemic," *Biomed. Eng. Comput. Biol.*, vol. 13, Jan. 2022, Art. no. 11795972221102115.
[18] H. Antil, "Mathematical opportunities in digital twins (MATH-DT)," 2024, *arXiv:2402.10326*.
[19] Ø. Bjelland, B. Rasheed, H. G. Schaathun, M. D. Pedersen, M. Steinert, A. I. Hellevik, and R. T. Bye, "Toward a digital twin for arthroscopic knee surgery: A systematic review," *IEEE Access*, vol. 10, pp. 45029–45052, 2022.
[20] T. Mîndra and A. M. Anghel, "Hybrid co-simulation framework for cyber-physical systems-based applications," in *Proc. Int. Conf. Appl. Math. Comput. Sci. (ICAMCS)*, Sep. 2024, pp. 102–109.
[21] S. Kumari and C. Prabha, "Role of digital twins and blockchain for the Internet of Medical Things," in *Blockchain and Digital Twin for Smart Hospitals*. Amsterdam, The Netherlands: Elsevier, Feb. 2025, pp. 67–87.
[22] Z. Yang, M. Chen, K. K. Wong, H. V. Poor, and S. Cui, "Federated learning for 6G: Applications, challenges, and opportunities," *Engineering*, vol. 8, pp. 33–41, Dec. 2021.
[23] Y. Zhao, J. Zhao, L. Jiang, R. Tan, D. Niyato, Z. Li, and Y. Liu,, "Privacy-preserving federated learning for healthcare: A blockchain-based approach," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 1817–1829, Aug. 2020.
[24] M. Aledhari, R. Razzak, R. M. Parizi, and F. Saeed, "Federated learning: A survey on enabling technologies, protocols, and applications," *IEEE Access*, vol. 8, pp. 140699–140725, Jul. 2020.
[25] L. Zhang, F. Zhou, and K. Liu, "Personalized healthcare using federated learning: A case study on cardiovascular disease," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 3, pp. 789–798, Jun. 2024.
[26] D. C. Nguyen, Q. V. Pham, P. N. Pathirana, M. Ding, A. Seneviratne, Z. Lin, and W. J. Hwang, "Federated learning for smart healthcare: A survey," *ACM Comput. Surv.*, vol. 55, no. 3, pp. 1–37, 2022.
[27] N. Nezhadsistani, N. S. Moayedian, and B. Stiller, "Blockchain-enabled federated learning in healthcare: Survey and state-of-the-art," *IEEE Access*, Jul. 2025.
[28] T. K. Dang, X. Lan, J. Weng, and M. Feng, "Federated learning for electronic health records," *ACM Trans. Intell. Syst. Technol.*, vol. 13, no. 5, pp. 1–17, May 2022.
[29] M. H. U. Rehman, W. H. L. Pinaya, P. Nachev, J. T. Teo, S. Ourselin, and M. J. Cardoso, "Federated learning for medical imaging radiology," *Brit. J. Radiol.*, vol. 96, no. 1150, Sep. 2023, Art. no. 20220890.
[30] S. A. Alzakari, A. Sarkar, M. Z. Khan, and A. A. Alhussan, "Converging technologies for health prediction and intrusion detection in Internet of Healthcare Things with matrix-valued neural coordinated federated intelligence," *IEEE Access*, vol. 12, pp. 99469–99498, Jun. 2024.
[31] S. Das, S. Dutta, S. Hazra, S. Nandi, A. Bandyopadhyay, and M. Disha, "Personalized healthcare empowered: Federated learning integration with wearable device data for enhanced patient insights," in *Proc. IEEE Region 10 Symp. (TENSYMP)*, Sep. 2024, pp. 1–6.
[32] Y. Li, S. Tan, and H. Wu, "Federated learning for predictive healthcare analytics: Trends and future directions," *IEEE Trans. Artif. Intell.*, vol. 4, no. 1, pp. 99–110, 2025.
[33] J. Elias, T. Ahmed, and L. Martins, "Robust federated learning for clinical decision support systems," *IEEE J. Sel. Topics Signal Process.*, vol. 18, no. 2, pp. 123–134, 2024.
[34] M. Singh, R. Srivastava, E. Fuenmayor, V. Kuts, Y. Qiao, N. Murray, and D. Devine, "Applications of digital twin across industries: A review," *Appl. Sci.*, vol. 12, no. 11, p. 5727, Jun. 2022.
[35] W. Kritzinger, M. Karner, G. Traar, J. Henjes, and W. Sihn, "Digital twin in manufacturing: A categorical literature review and classification," *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1016–1022, 2018.

[36] H. Hassani, X. Huang, and S. MacFeely, "Impactful digital twin in the healthcare revolution," *Big Data Cognit. Comput.*, vol. 6, no. 3, p. 83, Aug. 2022.
[37] L. De Oliveira El-Warrak and C. Miceli de Farias, "Could digital twins be the next revolution in healthcare?" *Eur. J. Public Health*, vol. 35, no. 1, pp. 19–25, Feb. 2025.
[38] M. A. Elaziz, M. A. A. Al-Qaness, A. Dahou, M. A. Al-Betar, M. M. Mohamed, M. El-Shinawi, A. Ali, and A. A. Ewees, "Digital twins in healthcare: Applications, technologies, simulations, and future trends," *WIREs Data Mining Knowl. Discovery*, vol. 14, no. 6, Nov. 2024, Art. no. e1559.
[39] A. Haleem, M. Javaid, R. P. Singh, and R. Suman, "Exploring the revolution in healthcare systems through the applications of digital twin technology," *Biomed. Technol.*, vol. 4, pp. 28–38, Dec. 2023.
[40] A. Vallée, "Envisioning the future of personalized medicine: Role and realities of digital twins," *J. Med. Internet Res.*, vol. 26, May 2024, Art. no. e50204.
[41] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108952–108971, 2020.
[42] Z. Yu, K. Wang, Z. Wan, S. Xie, and Z. Lv, "FMCPNN in digital twins smart healthcare," *IEEE Consum. Electron. Mag.*, vol. 12, no. 4, pp. 66–73, Jul. 2023.
[43] K. Bruynseels, F. S. de Sio, and J. van den Hoven, "Digital twins in health care: Ethical implications of an emerging engineering paradigm," *Frontiers Genet.*, vol. 9, p. 31, Feb. 2018.
[44] J. I. Jimenez, H. Jahankhani, and S. Kendzierskyj, "Health care in the cyberspace: Medical cyber-physical system and digital twin challenges," in *Digital Twin Technologies and Smart Cities*. Cham, Switzerland: Springer, 2020, pp. 79–92.
[45] M. Rahim, W. Lalouani, E. Toubal, and L. Emokpae, "A digital twin-based platform for medical cyber-physical systems," *IEEE Access*, vol. 12, pp. 174591–174607, 2024.
[46] L. Zhang, G. Yang, J. Li, and J. Liang, "A novel medical cyber-physical systems based on digital twin-driven platform," *Int. J. High Speed Electron. Syst.*, vol. 34, no. 4, Nov. 2024, Art. no. 2540129.
[47] A. Vallée, "Digital twin for healthcare systems," *Frontiers Digit. Health*, vol. 5, Sep. 2023, Art. no. 1253050.
[48] S. Rallapalli, M. R. Dileep, and A. V. Navaneeth, "Resource optimization with digital twins using intelligent techniques for smart healthcare management," in *Proc. Int. Conf. Inf.*, Jan. 2023, pp. 295–306.
[49] N. Wickramasinghe, P. P. Jayaraman, A. R. M. Forkan, N. Ulapane, R. Kaul, S. Vaughan, and J. Zelcer, "A vision for leveraging the concept of digital twins to support the provision of personalized cancer care," *IEEE Internet Comput.*, vol. 26, no. 5, pp. 17–24, Sep. 2022.
[50] M. N. Kamel Boulos and P. Zhang, "Digital twins: From personalised medicine to precision public health," *J. Personalized Med.*, vol. 11, no. 8, p. 745, Jul. 2021.
[51] M. Ayaz, M. F. Pasha, M. Y. Alzahrani, R. Budiarto, and D. Stiawan, "The fast health interoperability resources (FHIR) standard: Systematic literature review of implementations, applications, challenges and opportunities," *JMIR Med. Informat.*, vol. 9, no. 7, Jul. 2021, Art. no. e21929.
[52] W. K. Ozga, D. Zapała, P. Wierzgała, P. Augustynowicz, R. Porzak, and G. M. Wójcik, "Acoustic neurofeedback increases beta ERD during mental rotation task," *Appl. Psychophysiology Biofeedback*, vol. 44, no. 2, pp. 103–115, Jun. 2019.
[53] M. Bistroń and Z. Piotrowski, "Efficient video watermarking algorithm based on convolutional neural networks with entropy-based information mapper," *Entropy*, vol. 25, no. 2, p. 284, Feb. 2023.
[54] P. Lenarczyk and Z. Piotrowski, "Parallel blind digital image watermarking in spatial and frequency domains," *Telecommun. Syst.*, vol. 54, no. 3, pp. 287–303, Nov. 2013.
[55] P. Durneva, K. Cousins, and M. Chen, "The current state of research, challenges, and future research directions of blockchain technology in patient care: Systematic review," *J. Med. Internet Res.*, vol. 22, no. 7, Jul. 2020, Art. no. e18619.
[56] E. H. Park, H. I. Watson, F. V. Mehendale, A. Q. O'Neil, and C. Evaluators, "Evaluating the impact on clinical task efficiency of a natural language processing algorithm for searching medical documents: Prospective crossover study," *JMIR Med. Informat.*, vol. 10, no. 10, Oct. 2022, Art. no. e39616.
[57] Y. Ming and T. Zhang, "Efficient privacy-preserving access control scheme in electronic health records system," *Sensors*, vol. 18, no. 10, p. 3520, Oct. 2018.
[58] E. Entzeridou, E. Markopoulou, and V. Mollaki, "Public and physician's expectations and ethical concerns about electronic health record: Benefits outweigh risks except for information security," *Int. J. Med. Informat.*, vol. 110, pp. 98–107, Feb. 2018.
[59] T. Sondej, I. Jannasz, K. Sieczkowski, A. Dobrowolski, K. Obiała, T. Targowski, and R. Olszewski, "Validation of a new device for photoplethysmographic measurement of multi-site arterial pulse wave velocity," *Biocybernetics Biomed. Eng.*, vol. 41, no. 4, pp. 1664–1684, Oct. 2021.
[60] N. Jeyaraman, M. Jeyaraman, S. Yadav, S. Ramasubramanian, S. Balaji, S. Muthu, C. L. P, and B. P. Patro, "Applications of fog computing in healthcare," *Cureus*, vol. 16, no. 7, p. 64263, 2024.
[61] J. G. Carrasco Ramírez, "AI in healthcare: Revolutionizing patient care with predictive analytics and decision support systems," *J. Artif. Intell. Gen. Sci. (JAIGS) ISSN*, vol. 1, no. 1, pp. 31–37, Feb. 2024.
[62] A. Tacchino, J. Podda, V. Bergamaschi, L. Pedullà, and G. Brichetto, "Cognitive rehabilitation in multiple sclerosis: Three digital ingredients to address current and future priorities," *Frontiers Hum. Neurosci.*, vol. 17, Feb. 2023, Art. no. 1130231.
[63] C. Wu, G. Lorenzo, D. A. Hormuth, E. A. B. F. Lima, K. P. Slavkova, J. C. DiCarlo, J. Virostko, C. M. Phillips, D. Patt, C. Chung, and T. E. Yankeelov, "Integrating mechanism-based modeling with biomedical imaging to build practical digital twins for clinical oncology," *Biophys. Rev.*, vol. 3, no. 2, Jun. 2022, Art. no. 021304.
[64] C. Meijer, H.-W. Uh, and S. el Bouhaddani, "Digital twins in healthcare: Methodological challenges and opportunities," *J. Personalized Med.*, vol. 13, no. 10, p. 1522, Oct. 2023.
[65] R. Laubenbacher, F. Adler, G. An, F. Castiglione, S. Eubank, L. L. Fonseca, J. Glazier, T. Helikar, M. Jett-Tilton, D. Kirschner, P. Macklin, B. Mehrad, B. Moore, V. Pasour, I. Shmulevich, A. Smith, I. Voigt, T. E. Yankeelov, and T. Ziemssen, "Toward mechanistic medical digital twins: Some use cases in immunology," *Frontiers Digit. Health*, vol. 6, Mar. 2024, Art. no. 1349595.
[66] P. Jia, S. Liu, and S. Yang, "Innovations in public health surveillance for emerging infections," *Annu. Rev. Public Health*, vol. 44, no. 1, pp. 55–74, Apr. 2023.
[67] R.-P. Fischer, A. Volpert, P. Antonino, and T. D. Ahrens, "Digital patient twins for personalized therapeutics and pharmaceutical manufacturing," *Frontiers Digit. Health*, vol. 5, Jan. 2024, Art. no. 1302338.
[68] M. J. M. Cluitmans, G. Plank, and J. Heijman, "Digital twins for cardiac electrophysiology: State of the art and future challenges," *Herzschr Elektrophys.*, vol. 35, no. 2, pp. 118–123, 2024.
[69] P. Gazerani, "Intelligent digital twins for personalized migraine care," *J. Personalized Med.*, vol. 13, no. 8, p. 1255, Aug. 2023.
[70] T. Sigawi and Y. Ilan, "Using constrained-disorder principle-based systems to improve the performance of digital twins in biological systems," *Biomimetics*, vol. 8, no. 4, p. 359, Aug. 2023.
[71] IBM. *What is a Digital Twin*. Accessed: Feb. 4, 2025. [Online]. Available: https://www.ibm.com/topics/what-is-a-digital-twin
[72] M. Martínez-García and E. Hernández-Lemus, "Data integration challenges for machine learning in precision medicine," *Frontiers Med.*, vol. 8, Jan. 2022, Art. no. 784455.
[73] I. M. Putrama and P. Martinek, "Heterogeneous data integration: Challenges and opportunities," *Data Brief*, vol. 56, Oct. 2024, Art. no. 110853.
[74] N. Alhammad, M. Alajlani, A. Abd-Alrazaq, G. Epiphaniou, and T. Arvanitis, "Patients' perspectives on the data confidentiality, privacy, and security of mHealth apps: Systematic review," *J. Med. Internet Res.*, vol. 26, May 2024, Art. no. e50715.
[75] European Medicines Agency. *What We Do*. Accessed: Feb. 4, 2025. [Online]. Available: https://www.ema.europa.eu/en/about-us/what-we-do
[76] Internet Corporation for Assigned Names and Numbers. *ICANN for Beginners*. Accessed: Feb. 4, 2025. [Online]. Available: https://www.icann.org/en/beginners
[77] W. A. Woods, M. Watson, S. Ranaweera, G. Tajuria, and A. Sumathipala, "Under-representation of low and middle income countries (LMIC) in the research literature: Ethical issues arising from a survey of five leading medical journals: Have the trends changed?" *Global Public Health*, vol. 18, no. 1, Jan. 2023, Art. no. 2229890.
[78] M. N. Dinh, J. Nygate, V. H. M. Tu, and C. L. Thwaites, "New technologies to improve healthcare in low- and middle-income countries: Global Grand Challenges satellite event, Oxford University Clinical Research Unit, Ho Chi Minh City, 17th-18th September 2019," *Wellcome Open Res.*, vol. 5, p. 142, Jun. 2020.

[79] J. D. Sachs, "The lancet commission on lessons for the future from the COVID-19 pandemic," *Lancet*, vol. 400, no. 10359, pp. 1224–1280, 2022.
[80] C. H. Tsai, A. Eghdam, N. Davoody, G. Wright, S. Flowerday, and S. Koch, "Effects of electronic health record implementation and barriers to adoption and use: A scoping review and qualitative analysis of the content," *Life*, vol. 10, no. 12, p. 327, Dec. 2020.
[81] S. R. P. Franzen, C. Chandler, and T. Lang, "Health research capacity development in low and middle income countries: Reality or rhetoric? A systematic meta-narrative review of the qualitative literature," *BMJ Open*, vol. 7, no. 1, Jan. 2017, Art. no. e012332.
[82] C. S. Shumba and A. M. Lusambili, "Not enough traction: Barriers that aspiring researchers from low- and middle-income countries face in global health research," *J. Global Health Econ. Policy*, vol. 1, Jul. 2021, Art. no. 2021002.
[83] D. Drummond and A. Gonsard, "Definitions and characteristics of patient digital twins being developed for clinical use: Scoping review," *J. Med. Internet Res.*, vol. 26, Nov. 2024, Art. no. e58504.



HARSHIT SHARMA was born in Gohana, Haryana, India. He received the B.Tech. degree in computer science and engineering from Kurukshetra University, Haryana, in 2021, and the M.Tech. degree in computer science and engineering from Chandigarh University, Punjab, India, in 2023. He is currently pursuing the Ph.D. degree from MNIT Jaipur. After, he is an Assistant Professor with Manipal University Jaipur, Rajasthan, India. From 2022 to 2023, he was an Academic Researcher with Chandigarh University. In 2023, he was a Software Engineer Intern with Unisys Global Services, Bengaluru, India. His work has resulted in several publications in the fields of machine learning, healthcare applications, and cryptography, including research on diabetes prediction and hybrid cryptographic frameworks. He is a Certified Specialist in data science, machine learning, and natural language processing from IBM and Coursera. He is a UGC-NET qualified academic and was honored as the "Best Researcher" during his post-graduation. He has also held several academic coordination roles, including Course, Class, and Placement Coordinator. His research interests include supervised and unsupervised machine learning, deep learning, computer vision, and cryptographic systems.



SIMRAN KAUR was born in Patiala, Punjab, India. She received the B.Tech. degree in computer science and engineering from MRSPT University, Punjab, in 2021, and the M.Tech. degree in computer science and engineering from Chandigarh University, Punjab, in 2023.

She is currently an Assistant Professor with Manipal University Jaipur, Rajasthan, India. Prior to this, she was a Teaching Assistant with Chandigarh University, from 2022 to 2023, and as a Software Developer Intern with ClickLabs, Chandigarh, in 2020. She has co-authored multiple research articles focusing on swarm intelligence, PSO-ACO optimization, and healthcare-oriented image analysis. She has also contributed to academic development through her roles as a Course Coordinator, a Placement Coordinator, and a Departmental Event Coordinator. Her current research interests include medical image analysis, optimization algorithms, and intelligent healthcare systems.

Ms. Kaur has received several awards, including being recognized as the District Topper in Matriculation, and was awarded the Best Anchor in an IEEE event at Chandigarh University. She has been actively involved with the IEEE Photonics Society and served as a Faculty Representative.